

SQL & POWER BI BUSINESS ANALYTICS PROJECT

Superstore Analytics Report

Analyzing Superstore business data using SQL, an AI-powered natural language chatbot, and Power BI to generate insights and recommendations through interactive dashboards.

Power BI SQL AI Chatbot Business Analytics Data Visualization

TEAM MEMBERS

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DOCUMENT VERSION

1.0

Project Overview

A brief description of the project and the business objectives it addresses.

The Superstore Analytics Report project analyzes Superstore business data using SQL, an AI-powered natural language chatbot, and Power BI to generate meaningful insights and actionable recommendations. The project combines advanced SQL-based analysis and an AI Data Chatbot with a suite of interactive Power BI dashboards, giving stakeholders a clear, data-driven view of overall business, customer, employee, and territory performance.

Business Objectives

- ✓ Understand the business context and define clear analytical objectives
- ✓ Collect, prepare, and structure the Superstore dataset for analysis
- ✓ Use advanced SQL queries to uncover insights beyond standard reporting
- ✓ Build an AI-powered chatbot that lets stakeholders query the data warehouse in natural language
- ✓ Develop interactive Power BI dashboards covering Overview, Customer, Employee, and Territory perspectives
- ✓ Translate analytical findings into practical business recommendations

Business Understanding & Data Collection

This phase established the project's business context and prepared the dataset for analysis.

CONTRIBUTOR

Tarek Mostafa — Business Understanding & Data Collection

Responsibilities

Editable: Add specific details on data sources, business questions identified, or scope decisions made during this phase.

- ✓ Conducted business understanding to establish the context and purpose of the analysis
- ✓ Defined the project objectives and analytical requirements
- ✓ Collected and prepared the dataset for analysis, ensuring it was ready for SQL querying and dashboard development

AI Data Chatbot

An AI-powered assistant that lets stakeholders query the data warehouse in plain language and get instant business insights.

CONTRIBUTOR

Abdelrahman Reda — AI Data Chatbot

Overview

An AI-powered assistant that allows users to query the data warehouse using natural language and receive instant business insights.

Workflow

- User asks a business question.
- Gemini LLM generates the required SQL query.
- The query is validated and executed on the Data Warehouse.
- Results are converted into clear analytical insights.

Key Capabilities

- ✓ Natural Language → SQL Generation
- ✓ Star Schema Understanding
- ✓ Automated Data Analysis
- ✓ Business-Friendly Responses

Tech Stack

Python | Gemini AI | SQLite | Streamlit | SQLGlot

- ✓ Developed an AI-powered chatbot that translates natural language questions into SQL
- ✓ Generated advanced insights that could not be identified using standard dashboards alone
- ✓ Produced business recommendations grounded in the chatbot's automated analysis

Dashboard Development

Four interactive Power BI dashboards were developed, each focused on a distinct analytical perspective.

5.1 Overview Dashboard [EXECUTIVE SUMMARY]

DEVELOPED BY Nardin George

Responsibilities

- ✓ Designed and developed the Overview Dashboard
- ✓ Organized KPIs and executive-level visualizations, and created the dashboard layout and user-friendly design

Key Performance Indicators

Placeholder: Insert the KPIs featured on this dashboard (e.g. Total Sales, Profit Margin, Order Count).

Charts & Visualizations

Placeholder: Insert the charts and visuals used on this dashboard (e.g. bar charts, trend lines, maps).

[Screenshot Placeholder — Overview Dashboard — recommended 1600 × 900 px, PNG]

5.2 Customer Dashboard [CUSTOMER ANALYSIS]

DEVELOPED BY Asser Ayman

Responsibilities

- ✓ Designed and developed the Customer Dashboard
- ✓ Created customer-related KPIs and visualizations

Key Performance Indicators

Placeholder: Insert the KPIs featured on this dashboard (e.g. Total Sales, Profit Margin, Order Count).

Charts & Visualizations

Placeholder: Insert the charts and visuals used on this dashboard (e.g. bar charts, trend lines, maps).

[Screenshot Placeholder — Customer Dashboard — recommended 1600 × 900 px, PNG]

5.3 Employee Dashboard [EMPLOYEE ANALYSIS]

DEVELOPED BY Mohamed Reda

Responsibilities

- ✓ Designed and developed the Employee Dashboard
- ✓ Created employee performance KPIs and visualizations

Key Performance Indicators

Placeholder: Insert the KPIs featured on this dashboard (e.g. Total Sales, Profit Margin, Order Count).

Charts & Visualizations

Placeholder: Insert the charts and visuals used on this dashboard (e.g. bar charts, trend lines, maps).

[Screenshot Placeholder — Employee Dashboard — recommended 1600 × 900 px, PNG]

5.4 Territory Dashboard [TERRITORY ANALYSIS]

DEVELOPED BY Ahmed Zeater

Responsibilities

- ✓ Designed and developed the Territory Dashboard
- ✓ Created territory and regional performance visualizations

Key Performance Indicators

Placeholder: Insert the KPIs featured on this dashboard (e.g. Total Sales, Profit Margin, Order Count).

Charts & Visualizations

Placeholder: Insert the charts and visuals used on this dashboard (e.g. bar charts, trend lines, maps).

[Screenshot Placeholder — Territory Dashboard — recommended 1600 × 900 px, PNG]

Key Insights

Through analyzing the dashboards, we were able to identify several key insights across each analytical perspective.

Overview Dashboard

- ✓ The company is achieving strong overall performance in terms of both sales and profit.
- ✓ The profit margin is relatively high, which indicates that the company is generating healthy profits from its sales.
- ✓ California is the top-performing state and contributes a significant portion of the company's revenue.
- ✓ The number of orders fluctuates throughout the year and is not consistent across all periods.
- ✓ Profit is closely related to sales performance, meaning that increases in sales are generally accompanied by increases in profit.

Customer Dashboard

- ✓ One customer group contributes a larger share of revenue compared to the other groups.
- ✓ The company is not heavily dependent on a single customer, which helps reduce business risk.
- ✓ A small group of customers contributes a significant portion of overall profit and sales.
- ✓ Customer profitability varies considerably across different customer segments.
- ✓ Not all high-purchasing customers generate the highest profit margins.
- ✓ Certain customer groups demonstrate stronger purchasing behavior and higher business value.
- ✓ Customer concentration suggests dependence on a limited number of high-value customers.

Employee Dashboard

- ✓ Employee performance is uneven across the organization.
- ✓ A small number of employees contribute disproportionately to total sales and profit.
- ✓ High sales performance does not always correspond to high profitability.
- ✓ Performance differences indicate opportunities for knowledge sharing and skill development.
- ✓ Employee contribution can be clearly measured through sales volume, profit generation, and order quantity.

Territory Dashboard

- ✓ Geographic performance varies significantly between cities and states.
- ✓ Some territories achieve high sales volume while generating lower profit margins.
- ✓ A limited number of regions contribute a large share of overall business performance.
- ✓ Certain cities show strong profitability and may represent strategic growth opportunities.
- ✓ Underperforming territories indicate potential market expansion or operational challenges.

Recommendations

Based on these findings, we recommend the following actions across each analytical perspective.

Overview Dashboard

- ✓ Increase focus on lower-performing states to improve sales and reduce reliance on a single market.
- ✓ Analyze periods with lower order volumes to identify the reasons behind the decline and improve performance during those periods.
- ✓ Continue monitoring profit and profit margin alongside sales, as sales growth alone is not sufficient without maintaining profitability.

Customer Dashboard

- ✓ Develop strategies to attract and grow customer groups that currently contribute less revenue.
- ✓ Retain high-value customers through loyalty programs, personalized offers, or targeted engagement strategies.
- ✓ Apply successful practices from top-performing markets and customer groups to areas with lower performance.
- ✓ Develop loyalty programs targeting high-value customers to improve retention.
- ✓ Create personalized marketing campaigns for the most profitable customer segments.
- ✓ Identify low-profit customers with high purchase volume and optimize pricing strategies.
- ✓ Reduce dependency on a small customer base by expanding acquisition efforts toward similar profitable customer profiles.
- ✓ Implement customer segmentation strategies to improve targeting and resource allocation.

Employee Dashboard

- ✓ Analyze the practices of top-performing employees and share them across the team.
- ✓ Introduce performance-based incentive programs that reward profitability in addition to sales volume.
- ✓ Provide coaching and development plans for lower-performing employees.
- ✓ Establish regular performance reviews using dashboard KPIs.
- ✓ Balance workload and sales opportunities among employees to maximize overall productivity.

Territory Dashboard

- ✓ Increase investment and sales efforts in high-performing territories.
- ✓ Investigate factors affecting profitability in regions with strong sales but weak margins.
- ✓ Develop localized sales and marketing strategies based on regional performance patterns.
- ✓ Prioritize expansion initiatives in cities demonstrating strong profitability potential.
- ✓ Conduct deeper analysis of underperforming territories to identify operational or market-related issues.

Team Contributions Summary

Summary of each team member's role and responsibilities across the project.

Team Member	Role	Responsibilities
Tarek Mostafa	Business Understanding & Data Collection	- Conducted business understanding - Defined project objectives and requirements - Collected and prepared the dataset for analysis
Abdelrahman Reda	AI Data Chatbot	- Built the natural language → SQL chatbot on Gemini AI - Implemented query validation and star-schema understanding - Produced business insights from automated data analysis
Nardin George	Overview Dashboard Developer	- Designed and developed the Overview Dashboard - Organized KPIs and executive-level visualizations - Created the dashboard layout and user-friendly design
Asser Ayman	Customer Dashboard Developer	- Designed and developed the Customer Dashboard - Created customer-related KPIs and visualizations
Mohamed Reda	Employee Dashboard Developer	- Designed and developed the Employee Dashboard - Created employee performance KPIs and visualizations
Ahmed Zeater	Territory Dashboard Developer	- Designed and developed the Territory Dashboard - Created territory and regional performance visualizations

Conclusion

The Superstore Analytics Report project demonstrates how combining advanced SQL analysis, an AI-powered data chatbot, and interactive Power BI dashboards can turn raw business data into clear, actionable insight. Through structured business understanding, in-depth SQL querying, a natural-language chatbot for on-demand insights, and purposeful dashboard design across Overview, Customer, Employee, and Territory perspectives, the project delivers a comprehensive analytical view of Superstore's performance.

The insights and recommendations produced through this work provide a solid foundation for data-driven decision-making, and the dashboard and chatbot suite offers a reusable toolset for ongoing performance monitoring.

“Turning Superstore data into clear insight and actionable recommendations — through advanced analysis, an AI-powered chatbot, and interactive dashboards.”

PREPARED BY

Tarek Mostafa, Abdelrahman Reda, Nardin George, Asser Ayman, Mohamed Reda, Ahmed Zeater

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